

Distributed Audio DSP Middleware — Milestone 2 Update

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Status: Milestone 2 complete. The system now performs a full remote audio round-trip with live return: the plugin streams a track's audio to a peer machine, the peer processes it, and the processed audio is returned over the network and played back in the DAW in real time, time-aligned with the rest of the session. This closes the loop on Milestone 1's one-way transport.

What was completed this milestone

- **Send/return plugin.** The VST3 plugin is now bidirectional — it streams dry audio out and plays the processed result back in place of the dry signal. Real-time safety extends to both directions: no socket I/O ever runs on the DAW audio thread.
- **Fixed-latency playback (host delay compensation).** The plugin declares a fixed latency to the DAW and outputs the processed version of audio it sent earlier; the host time-aligns the track, making remote processing behave like a normal insert effect. This is the core technical result of the milestone.
- **Custom wire protocol v2.** Added an absolute timeline stamp (so returned packets land at the correct output sample regardless of arrival order) and a direction flag, over dual UDP ports (dry out / processed back).
- **New real-time-safe components.** A lock-free receive buffer and a dry-signal delay line, plus a dedicated network receive thread — mirroring the Milestone 1 sender design.
- **Graceful degradation.** If a processed packet arrives late or is lost, the plugin outputs the original dry audio for that instant (effectively bypassed) and counts it, rather than producing an audible dropout.
- **Remote processing node.** A lightweight processing node (Python, standard library) applies a built-in DSP effect (gain + low-pass filter) to demonstrate genuine remote processing and return.
- **Validation.** Verified first on loopback, then on a real two-machine LAN (laptop host in Ableton Live → PC processing node): processed audio returns and plays time-aligned, with packet and underrun metrics captured.

Changes to scope

- **Milestone split (deliberate).** Milestone 2 was originally framed to also host real third-party VST3 plugins on the remote node. Given a compressed timeline (full-time co-op plus a concurrent course), the milestone was scoped to the transport and live-return engine, proven with a built-in DSP node, and real VST3 hosting was moved to Milestone 3. This de-risks the schedule by separating the now-completed real-time/distributed transport from the higher-variance plugin-hosting work.
- **Stretch goals not attempted.** Optional stretch goals (a C++/JUCE node application and hosting a single hardcoded VST3) were initially considered but intentionally deferred to Milestone 3 - where they fit naturally.
- **Architecture direction confirmed.** A peer-to-peer, LAN-first approach (rather than cloud-first) to mitigate round-trip latency, jitter, and plugin license verification — consistent with prior discussion.

Updated plan

- **Milestone 3 — Remote VST3 hosting service.** A C++/JUCE processing-node application that hosts real, selectable third-party VST3 plugins; a reliable TCP control channel for plugin selection and live parameter control; a parameter UI that scales with the chosen plugin; and automatic LAN peer discovery (the latter droppable to future work, since manual configuration already functions).
- **Milestone 3+ (planned) — Offline freeze/flatten.** Capture a pass, process it losslessly on the node, and cache the result locally for network-free playback — a “freeze” workflow over the reliable TCP tier.
- **Beyond.** Cloud tier with license-aware offloading; fault tolerance via modified Raft on the control plane with volume-crossfade failover.

Milestone 2 deliverables (plan, implementation, presentation) are committed to the project repository and filed under Documents/. Known, documented limitations this milestone: linear-playback timeline assumption; built-in DSP node (real plugin hosting is Milestone 3); single channel pair.